INVENTORY MANAGEMENT SYSTEM

Inventory Management System — Project Report

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Executive Summary

- Figures stated exactly as requested: 54,200.00.

A professional project report for a small inventory management system built with Python, Flask, SQLite, and a vanilla HTML/CSS/JavaScript front end.

Objectives

To develop an efficient inventory management system that can handle multiple products, track stock levels, and provide real-time updates to the frontend.

System Architecture

The system is designed with a RESTful API using Flask for backend operations and SQLite as the database. The frontend is built with vanilla HTML/CSS/JavaScript, providing a clean interface for users to interact with the inventory.

Database Schema

The database schema includes tables for products, stock levels, and user profiles (if implemented).

Product Name	Quantity	Unit Price
Keyboard	5	120
Mouse	3	40
Monitor	8	200
Laptop Stand	2	60
USB Hub	10	15

JSON API Endpoints

The system provides endpoints for adding products, updating stock levels, and retrieving inventory data in JSON format.

Test Results

32 automated unit tests passed and 19 browser checks passed.

Limitations

Single-user profiles are not verified. No authentication is implemented, and the generic events table is used for tracking inventory changes.

Future Improvements

Implement user authentication to secure access to the system. Add more features such as product categories, reorder levels, and notifications for low stock.